

1. PRODUCT AND COMPANY IDENTIFICATION

Product Name: IP-XXX
Product Codes: 413-XXX
Synonyms: Copper Infiltrating Powder
Emergency: Contact: CHEMTREC

SDS Number: IPA
Product Use: For Powder Metallurgy Applications.
Restrictions: Industrial Use Only.
Manufacturer: SCM Metal Products, Inc.

For U.S., Canada, Puerto Rico, & U.S. Virgin Islands:
(703) 741-5970 / (800) 424-9300

Outside North America:
See Section 16

SCM Metal Products, Inc.
2601 Weck Drive, Box 12166
Research Triangle Park, NC 27709-2166
(919) 544-8090

2. HAZARDS IDENTIFICATION**Health Hazards:**

None Known

Environmental Hazards:

Acute Aquatic Toxicity – Category 1
Chronic Aquatic Toxicity – Category 3

Pictogram:**Physical Hazards:**

None Known

Signal Word: Warning

Hazard Statements:

H400: Very toxic to aquatic life.
H412: Harmful to aquatic life with long lasting effects.

Precautionary Statements:

P273: Avoid release to the environment.
P391: Collect spillage.
P501: Dispose of contents/container in accordance with local/regional/national/international regulations.

Other Hazards:

Inhalation of dusts or fumes leads to irritation of respiratory system. Inhalation of higher concentrations may cause metal fume fever.

3. COMPOSITION/INFORMATION ON INGREDIENTS

<u>Component</u>	<u>CAS Number</u>	<u>EINECS Number</u>	<u>Range % by Wt.</u>
Copper	7440-50-8	231-159-6	90 – 98
Iron	7439-89-6	231-096-4	0.5 – 6
Zinc	7440-66-6	231-175-3	0.25 – 0.70
Silicon	7440-21-3	231-130-8	0.02 – 0.10
Manganese	7439-96-5	231-105-1	< 3.0

Note: A portion of the copper, iron, silicon, and zinc come from a master mix alloy.

4. FIRST AID MEASURES**EYES:**

Flush eyes with plenty of water, lifting the upper and lower eyelids occasionally. Get medical attention if irritation develops.

SKIN:

Wash the skin using soap or a mild detergent and warm water.

INHALATION:

Move the exposed person to fresh air at once. If breathing has stopped, perform artificial respiration. Get immediate medical attention. Fume from metallizing, welding or similar processes can cause respiratory irritation and/or metal fume fever (respiratory irritation, chills, nausea).

INGESTION:

If person is conscious, rinse mouth and give large quantities of water to drink. Get medical attention.

5. FIRE FIGHTING MEASURES**EXTINGUISHING MEDIA:**

Graphite, dolomite or sodium chloride. Do NOT use water.

UNUSUAL FIRE AND EXPLOSION HAZARDS:

Copper powder with particles sizes 50µ size range are classified as weakly explosive by the U.S. Bureau of Mines Report RI-6516. When present as a dust cloud, will NOT explode readily in air. Not easily ignited by sparks.

FIRE FIGHTING EQUIPMENT:

Wear full bunker gear including a positive pressure self-contained breathing apparatus.

PRECAUTIONS:

Keep away from ignition sources (e.g. heat and open flames). None required. Keep container closed.

HAZARDOUS DECOMPOSITION:

None identified.

6. ACCIDENTAL RELEASE MEASURES

1. Restrict the area to those persons wearing respiratory protection. Do not allow unprotected people into the area until cleanup has been completed.
2. Ventilate the area thoroughly.
3. Collect the powder in a manner that minimizes further dust generation.
4. Keep out of sewers and waterways.
5. Recycle or dispose of as a waste (see Section 13).

7. HANDLING AND STORAGE

Avoid dust generation. Wash thoroughly after handling. Eating, drinking, and smoking are prohibited in work areas. Store powder in a dry area, -18° to 44°C.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION***Ventilation Requirements:***

Keep dust and fume levels below occupational exposure limits. Provide adequate local exhaust ventilation at the work area where spraying and fusing are being done.

Personal Protective Equipment:**EYES:**

Wear dust-proof safety goggles. Contact lenses are not recommended.

SKIN:

None required; however, use of protective gloves and clothing is good industrial practice. The use of impervious gloves or barrier cream to protect the skin is recommended.

INHALATION:

Do not breathe dust or fume. Use with adequate ventilation. Use NIOSH/MSHA approved respirator.

OCCUPATIONAL EXPOSURE LIMITS:

Copper Dust and Mists

CAS# 7440-50-8

EINECS# 231-159-6

ACGIH TLV	1.0 mg/m ³ TWA
NIOSH REL	1.0 mg/m ³ TWA
* NIOSH IDLH	100 mg/m ³
OSHA PEL	1.0 mg/m ³ TWA

IDLH = Immediately dangerous to life and health.*Copper Fume**

ACGIH TLV	0.2 mg/m ³ TWA
NIOSH REL	0.1 mg/m ³ TWA
* NIOSH IDLH	100 mg/m ³
OSHA PEL	0.1 mg/m ³ TWA

IDLH = Immediately dangerous to life and health.*Iron**

CAS# 7439-89-6

EINECS# 231-096-4

ACGIH TLV	0.2 mg/m ³ TWA
NIOSH REL	0.1 mg/m ³ TWA
* NIOSH IDLH	100 mg/m ³
OSHA PEL	0.1 mg/m ³ TWA

IDLH = Immediately dangerous to life and health.*Zinc**

CAS# 7440-66-6

EINECS# 231-175-3

ACGIH TLV	1.0 mg/m ³ TWA; 2.0 mg/m ³ STEL
NIOSH REL	1.0 mg/m ³ TWA; 2.0 mg/m ³ STEL
* NIOSH IDLH	Effective IDLH 2000 mg/m ³
OSHA PEL	1.0 mg/m ³ TWA; 2.0 mg/m ³ STEL

IDLH = Immediately dangerous to life and health.*Silicon**

CAS# 7440-21-3

EINECS# 231-130-8

ACGIH TLV	N.D.
NIOSH REL	10 mg/m ³ (total) TWA; 5 mg/m ³ (resp)
* NIOSH IDLH	N.D.
OSHA PEL	15 mg/m ³ (total) TWA; 5 mg/m ³ (resp)

IDLH = Immediately dangerous to life and health.*Manganese**

CAS# 7439-96-5

EINECS# 231-105-1

ACGIH TLV	0.02 mg/m ³ TWA (resp.); 0.01 mg/m ³ TWA (Inhal.)
NIOSH REL	1.0 mg/m ³ TWA; 3.0 mg/m ³ ST
* NIOSH IDLH	500 mg/m ³
OSHA PEL	5.0 mg/m ³ C

***IDLH = Immediately dangerous to life and health.**

Revised IDLH: 500 mg Mn/m³
Basis for revised IDLH: The revised IDLH for manganese is 500 mg Mn/m³ based on acute inhalation toxicity data in animals [Gupta et al. 1981]. This may be a conservative value due to the lack of relevant acute toxicity data for workers.

9. CHEMICAL AND PHYSICAL PROPERTIES

APPEARANCE AND ODOR	Dull greyish powder; odorless
FLASH POINT	Above 700°C
FLAMMABILITY	Non-flammable
AUTOIGNITION TEMPERATURE	Not determined
pH	Not applicable
VAPOR PRESSURE	1mm HG@ 1628°C
VAPOR DENSITY	Not determined
MELTING POINT	1083°C
BOILING POINT	2595°C
SOLUBILITY IN WATER	Not soluble
SOLUBILITY IN FAT	Not determined
OCTANOL/WATER PARTITION COEFFICIENT	Not determined
RELATIVE DENSITY (WATER=1)	8.2
VISCOSITY	Not applicable

10. STABILITY AND REACTIVITY**STABILITY:**

Stable to ignition temperature of 700°C.

INCOMPATIBLE MATERIALS:

Copper is explosively incompatible with sodium azide. Copper dusts may react with acetylene gas to form copper acetylides, which are sensitive to shock. Copper is incompatible with oxidizers and alkalis. Copper mists may react with magnesium to form flammable hydrogen gas.

HAZARDOUS DECOMPOSITION:

None identified.

HAZARDOUS POLYMERIZATION:

Will not occur.

11. TOXICOLOGICAL INFORMATION

Copper is an essential element of mammalian metabolism. Copper metal has little or no serious toxicity. The most common adverse effect associated with copper is the acute inhalation of copper fume during refining or welding. Inhalation of copper fume or dust may result in metal fume fever, which is characterized by upper respiratory irritation, chills, metallic or sweet taste, nausea, and aching muscles. Attacks usually begin after 4-8 hours of exposure and last only 24-48 hours. Inhalation of fumes has been reported to sometimes cause discoloration of the skin and hair. Nausea and vomiting may result if larger amounts of copper metal are ingested. This is probably due to the conversion of the swallowed metal copper to its irritating salts. It is unlikely that poisoning by ingestion in industry would progress to a serious point because small amounts induce vomiting, emptying the stomach of copper salts. High airborne concentrations of copper metal would be expected to cause mechanical irritation of the eyes and respiratory tract. Metallic copper may cause keratinization of the hands and soles of the feet, but it is not commonly associated with industrial dermatitis.

Manganese intake is necessary for human health at low levels in the body; however, exposure to high concentrations may be toxic. Inhalation is the primary route of exposure for manganese in occupational settings. Manganism is a permanent neurological disorder that is a result of manganese toxicity. Symptoms such as irritability, aggressiveness, hallucinations, tremors, difficulty walking, and facial spasms may be experienced with manganism. Studies suggest that overexposure to manganese in the workplace may lead to adverse cognitive effects such as difficulty with concentration and memory problems. Manganese dust can cause mild to moderate irritation to the eyes, nose, and throat. Symptoms of eye and skin irritation may be experienced depending on the amount of exposure. Inhalation of manganese fume or dust may result in metal fume fever, which is characterized by upper respiratory irritation, chills, metallic or sweet taste, nausea, and aching muscles. In severe cases, lung toxicity from inhaled manganese may lead to an increased susceptibility to infections such as bronchitis, and may result in manganic pneumonia. There is insufficient evidence to suggest that manganese is carcinogenic. The IRIS has provided manganese with a weight-of-evidence Classification D – not classifiable as to human carcinogenicity.

No component of this product present at levels greater than 0.1% is identified as a carcinogen by the U.S. National Toxicology Program, the U.S. Occupational Safety and Health Act, or the International Agency for Research on Cancer (IARC).

12. ECOLOGICAL INFORMATION

No data on the ecological effects of this product have been developed.

13. DISPOSAL CONSIDERATIONS

Disposal must be in accordance with applicable local, state and federal regulations (contact local, state, or federal environmental agency for specific rules). Do not dump into sewers, on the ground, or into any body of water.

14. TRANSPORTATION INFORMATION

DOT: RQ, UN3077, Environmentally Hazardous Substance, Solid NOS (contains Copper), 9, III Marine Pollutant.
 DOT EXCEPTION: *Under 49 CFR 171.4, except when transporting aboard a vessel, the requirements of this subchapter specific to marine pollutants do not apply to non-bulk packaging transported by motor vehicles, rail cars, and aircraft.*
 ADR/RID: UN3077, Environmentally Hazardous Substances, Solid, NOS (contains Copper), 9, III Marine Pollutant.
 IMO/IMDG: UN3077, Environmentally Hazardous Substances, Solid, NOS (contains Copper), 9, III Marine Pollutant.
 ICAO/IATA: UN3077, Environmentally Hazardous Substances, Solid, NOS (contains Copper), 9, III Marine Pollutant.
 REPORTABLE QUANTITY: *Copper 5,000 lbs.*
 ADDITIONAL REMARKS: *Except when all or part of the transportation is by vessel, the above requirements do not apply to non-bulk packaging (<882 lbs.) transported by motor vehicle, rail car or aircraft.*

15. HAZARDOUS MATERIAL IDENTIFICATION SYSTEM/REGULATORY INFORMATION

Health Hazard: I – Slight: Slightly Toxic – May cause slight irritation.
 Flammability Hazard: 0 – Minimal: Will not burn under normal conditions.
 Reactivity Hazard: 0 – Minimal: Normally stable, does not react with water.
 Minimum Personal Protection: E – Safety Glasses, Gloves & Dust Respirator.

Copper and Manganese are on the Sara Title III, Section 313 Toxic Chemicals List.

Copper, iron, and manganese are listed on the following inventories (non-exhaustive):

	EU EC Inventory	USA TSCA	Canada DSL	Australia AOCs	New Zealand NZIoC	Japan ENCS	Korea ECL	Philippines PICCS	China IECSC
Copper	YES	YES	YES	YES	YES	YES	YES	YES	YES
Iron	YES	YES	YES	YES	YES	NO	YES	YES	YES
Zinc	YES	YES	YES	YES	YES	YES	YES	YES	YES
Silicon	YES	YES	YES	YES	YES	YES	YES	YES	YES
Manganese	YES	YES	YES	YES	YES	NO	YES	YES	YES

16. OTHER INFORMATION

Revision: D

May 12, 2021

Format has been updated to meet the new OSHA Hazard Communication Standard (1/27/2015). Sections 1-3 have been updated (8/4/2015). Updated Sections 1, 3, 8, 9, 14, 15, 16, and the Header & Footer; Rev. C (4/9/20). Updated Sections 3, 8, and 15 (5/12/2021).

See Emergency Telephone Numbers Below:

Emergency Numbers Outside of North America

Country	Greeting Language	Number Type	City	Region	Local Number	Toll-Free Number
Argentina	Latin American Spanish	Local (City)	Buenos Aires	Americas	54-1159839431	
Australia	English	Local (City)	Sydney	Oceania	61-290372994	
Austria	Austro- Bavarian / German	Local (City)	Vienna	Europe	43-13649237	
Belgium	French, Flemish, and German	Local (City)	Brussels	Europe	32-28083237	
Brazil	Portuguese	Local (City)	Rio De Janeiro	Americas	55-2139581449	
Brazil	Portuguese	Local (City)	Sao Paulo	Americas	55-1143491359	
Brazil - Toll Free	Portuguese	Toll Free Mobile-Enabled		Americas	0800 892 0479	0800 892 0479
Bulgaria	Bulgarian	Local (City)	Plovdiv	Europe	359-32570104	
Cayman Islands	English	Local (National)	Local (National)	Americas	345-749-8392	
Chile	Latin American Spanish	Local (City)	Santiago	Americas	56 2 2581 4934	
China	Mandarin	Local (National)		Asia	4001-204937	
Colombia	Latin American Spanish	Toll Free		Americas	01800-710-2151	01800-710-2151
Costa Rica	Latin American Spanish	Local (National)		Americas	506-40003869	
Croatia	Croatian	Local (City)	Zagreb	Europe	385-17776920	
Czech Republic	Czech	Local (City)	Prague	Europe	420-228880039	
Denmark	Danish	Local (National)		Europe	45-69918573	
Dominican Republic	Latin American Spanish	Local (City)	Santo Domingo	Americas	1 (829) 956-7588	
El Salvador	Latin American Spanish	Local (City)	San Salvador	Americas	503 2136 7633	
Estonia	Estonian	Local (National)		Europe	372-6681294	
Finland	Finnish	Local (City)	Helsinki	Europe	358-942419014	
France	French	National		Europe	33-975181407	
Germany	German	Toll Free Mobile-Enabled;#Local (City)	Frankfurt	Europe	49-69643508409	0800-181-7059
Greece	Greek	Local (City)	Athens	Europe	30-2111768478	
Grenada	English	Local (City)	St George	Americas	1 (473) 230-0165	
Guinea	French	Toll Free Mobile-Enabled		Americas	224 660 71 03 00	
Hong Kong	Cantonese	Toll Free		Asia	800-968-793	800-968-793
Hungary	Hungarian	Local (City)	Budapest	Europe	36-18088425	
Iceland	Icelandic	Local (City)	Reykjavik	Europe	354 539 0655	
India	Hindi, Bengali, English	Toll Free		Asia	000-800-100-7141	000-800-100-7141
Indonesia	Indonesian	Toll Free		Asia	001-803-017-9114	001-803-017-9114
Ireland	English	Local (City)	Dublin	Europe	353-19014670	
Israel	Hebrew	Local (City)	Tel Aviv	Asia	972-37630639	
Italy	Italian	Toll Free Mobile-Enabled;#Local (City)	Milan	Europe	39-0245557031	800-789-767
Japan	Japanese	Local (City)	Tokyo	Asia	81-345209637	
Latvia	Latvian/Lettish and Possibly Russian	Local (City)	Riga	Europe	371-66165504	
Lithuania	Lithuanian	Local (City)	Vilnius	Europe	370-52140238	
Luxembourg	Luxembourgish, French and German	Local (National)		Europe	352-20202416	
Macedonia	Macedonian	Local (City)	Skopje	Europe	389 2 551 7456	
Malaysia	Malay	Toll Free;#Local (City)	Kuala Lumpur	Asia	60-392125794	1-800-815-308
Mexico	Latin American Spanish	Toll Free		Americas	01-800-681-9531	01-800-681-9531
Netherlands	Dutch	Local (National)		Europe	31-858880596	
New Zealand	English	Local (City)	Auckland	Oceania	64-98010034	
Nigeria	Hausa	Local (City)	Lagos	Europe	234 1 227 8883	
Panama	Latin American Spanish	Local (National)		Americas	507-8322475	
Peru	Latin American Spanish	Local (City)	Lima	Americas	51-17071295	
Philippines	Tagalog	Toll Free;#Local (City)	Manila	Asia	+63 2 8395 3308 and 1-800-1-116-1020	1-800-1-116-1020
Poland	Polish	Local (City)	Warsaw	Europe	48-223988029	
Portugal	Portuguese	Local (National)		Europe	351-308801773	
Romania	Romanian	Local (National)		Europe	40-37-6300026	
Russia	Russian	Toll Free Mobile-Enabled		Asia	8-800-100-6346	8-800-100-6346
Saudi Arabia	Arabic and English	CHEMTREC DID Only		Asia	966-8111095861	
Singapore	English and Mandarin	Toll Free;#National		Asia	65-31581349	800-101-2201
Slovakia	Slovak	Local (City)	Bratislava	Europe	421-233057972	
Slovenia	Slovene/Slovenian	Local (City)	Ljubljana	Europe	38-618888016	
South Africa	English	Toll Free		Africa	0-800-983-611	0-800-983-611
South Korea	Korean	Toll Free		Asia		003-0813-2549 and 080-822-1374
Spain	European Spanish	Toll Free Mobile-Enabled;#Local (City)	Barcelona	Europe	34-931768545	900-868538
Sweden	Swedish	Local (City)	Stockholm	Europe	46-852503403	
Switzerland	German and French and Italian	Local (City)	Zurich	Europe	41-435082011	
Taiwan	Mandarin	Toll Free;#Local (City)	Taipei	Asia	886-2-7741-4207	00801-14-8954
Thailand	Thai	Toll Free		Asia	001-800-13-203-9987	001-800-13-203-9987
Trinidad and Tobago	English	Local (National)	National Number	Americas	1-868-224-5716	
Turkey	Turkish	Local (City)	Istanbul	Europe	90-212-7055340	
Ukraine	Ukrainian	Ukraine Mobile Number		Europe	380-947101374	
United Kingdom	English	Local (National);#Local (City)	London	Europe	44-870-8200418 and 44-2038073798	

The information in this SDS relates to this specific product group. It may not be valid for this product if used in combination with any other materials or in any process. It is the user's responsibility to satisfy himself as to the suitability and completeness of this information for his own particular use.